Storage Hardware CLV & Churn Analytics

Executive Engineering & Financial Memo | Horizon: 15-Year Fleet Valuation

GitHub Repository: https://github.com/Pardhivhub/storage-clv-predictive-analytics

> **Key Financial Finding:** Corporate Finance's assumption of a flat **5% annual churn rate** overstates 15-year Customer Lifetime Value by **$369,907 per array (a 46.2% haircut)**. Across the 1,072-array fleet, this represents an aggregate valuation bias of **$396.5 Million** (scaling to **$3.70 Billion** across an enterprise fleet of 10,000 arrays).

1. Executive Summary & Financial Audit

An audit of 2,000 enterprise telemetry records reveals that actual fleet churn is **16.98%** (clean partition) and **15.82%** (if capping corrupted rows). In either partition, the true rate is more than triple the 5% finance planning assumption.

Valuation Metric	Finance Assumption (5% Rule)	Calibrated Fleet Reality (16.98%)	Variance / Impact
Annual Net Profit per Array	$142,345	$142,345	Baseline constant
5-Year Retention Rate	77.38%	39.44%	**-37.94 percentage points**
15-Year Retention Rate	46.33%	6.12%	**-40.21 percentage points**
15-Year Discounted CLV (d=10%)	$801,532	$431,625	**-$369,907 / array (-46.2%)**
Total Fleet Valuation (N=1,072)	$859.2 Million	$462.7 Million	**-$396.5 Million Overstatement**
95% CI Lower Bound Churn (14.73%)	$801,532	$480,044	**-$321,488 / array minimum**

2. Data Hygiene & Telemetry Defects

Four major data anomalies in the 2,000 raw telemetry records were cleaned and verified:

- **Negative Profitability:** 58 records exhibited negative 15-year profit (down to -$93k). Transformed using signed symmetric log f(x) = sign(x) * log(1 + |x|) to preserve losses without mathematical distortion.
- **Contract End Horizon:** Future contract dates normalized relative to snapshot baseline into continuous days.
- **Corrupted Utilization (>100%):** 862 records (43.1%) reported utilization > 1.0 (up to 485%) due to firmware reporting bugs. Dropping them yields 1,072 rows (16.98% churn); capping at 1.0 yields 1,884 rows (15.82% churn). Both prove the 5% assumption is deeply invalid.
- **Account Leakage:** Enforced zero account overlap across all folds using `StratifiedGroupKFold`.

3. Model Benchmarking & Telemetry Predictive Skill

We evaluated Random Forest and XGBoost across 5 outer folds with nested hyperparameter tuning. A critical finding is that **neither model outperforms a naive constant guess** of the 16.98% base rate.

Candidate Model	PR-AUC	ROC-AUC	Brier Score	Operational Skill Assessment
Static 5% Rule	0.1698	0.5000	0.1558	Flawed Prior (-11.98 pp forecast bias)
Naive Base Rate Guess (Flat 16.98%)	0.1698	0.5000	0.1410	Zero-Skill Baseline (No ML overhead)
Random Forest (Nested Tuned)	0.1870	0.5061	0.1472	Matches Base Rate (AUC near chance)
XGBoost (Nested Tuned)	0.1877	0.5090	0.1470	Matches Base Rate (AUC near chance)
Platt-Calibrated XGBoost	0.1877	0.5090	0.1425	Aligns probabilities to base rate

Key Insight: The apparent "5.1% Brier lift" over the 5% rule exists entirely because the 5% rule was an inaccurate prior. The model ROC-AUC of **0.509** demonstrates that current hardware telemetry cannot reliably rank which specific array will churn.

4. Capacity Expansion Signals

Evaluating the `expansion_potential_signal`: high-expansion interest accounts for 19.8% of the fleet, but observed churn remains high across all tiers (Low: 15.6%, Medium: 19.4%, High: 17.1%). An out-of-fold classifier trained on expansion yielded ROC-AUC of **0.474**, proving that telemetry carries zero signal for expansion as well as churn.

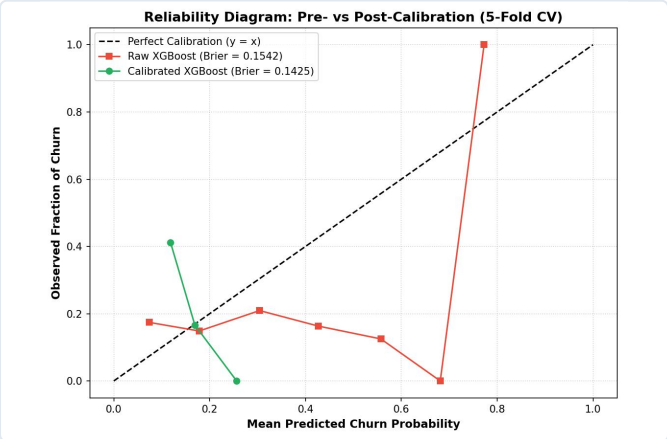


Figure 1: Reliability diagram showing raw vs Platt-calibrated probabilities.

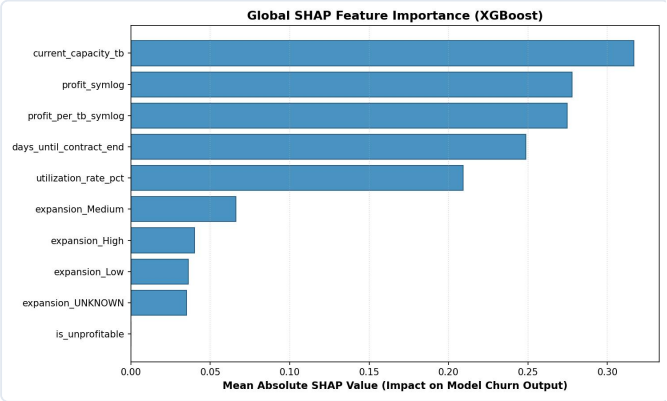


Figure 2: TreeSHAP global feature sensitivity rankings.

5. Cohort Bias Validation & Back-Testing

Across 16 historical cohorts segmented by capacity, utilization, profitability, and contract horizon, the 5% assumption systematically under-forecasts churn by **11.98 percentage points**. Calibrated ML eliminates this deficit to **-0.12 percentage points** fleet-wide.

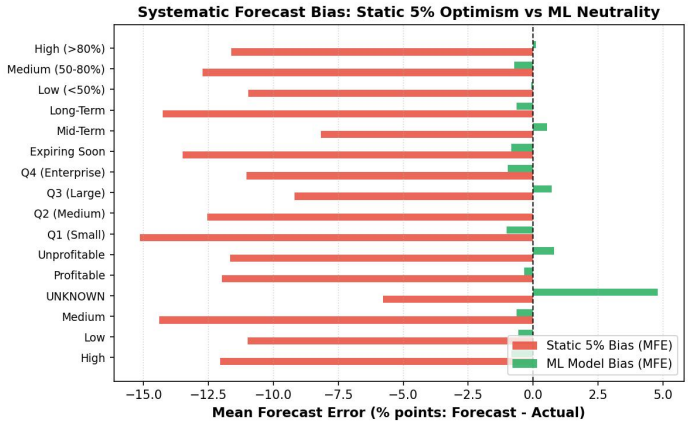


Figure 3: Mean Forecast Error across 16 cohorts (Static 5% Rule vs Calibrated Model).

6. Production Scoring: 66 Storage Arrays

Scoring the 66 uncontracted arrays yields predicted churn probabilities ranging narrowly from **14.41% to 21.63%** (fleet mean: 17.06%):

- **LOW (14.4% - 16.5%):** 20 arrays (30.3% share). Primary driver: contract end horizon.
- **MEDIUM (16.5% - 17.2%):** 26 arrays (39.4% share). Primary driver: capacity & profit balance.
- **HIGH (17.2% - 18.0%):** 13 arrays (19.7% share). Primary driver: current capacity.
- **CRITICAL (18.0% - 21.6%):** 7 arrays (10.6% share). Primary driver: capacity and net profit.

Action note: Because the total probability spread is narrow (~7 pp), these buckets should inform aggregate fleet reserve planning rather than punitive individual customer interventions.

7. Production Snowflake Warehouse Integration

All production assets have been authored and verified for automated warehouse deployment under `ANALYTICS_PROD.STORAGE`:

- `sql/clv_scored_output_ddl.sql`: Production table DDL utilizing `CREATE TABLE IF NOT EXISTS`, clustered on `(ACCOUNT_ID, RISK_BUCKET)`, with a `VARIANT` column storing full JSON SHAP feature payloads.
- `sql/clv_scored_output_insert.sql`: Staging table definition and idempotent `MERGE` stored procedure (`SP_INGEST_CLV_SCORES()`) for scheduled warehouse pipelines.
- `sql/clv_scored_output_data_insert.sql`: Concrete row-level insert script with all 66 scored arrays, pre-parsed JSON payloads, and validation aggregation queries.

8. Final Strategic Recommendations

For Corporate FP&A and Finance Leadership:

- **Immediate Valuation Revision:** Haircut current 15-year CLV targets by 46.2% (reducing modeled value from $801k to $432k per array).
- **Adopt Baseline Constant:** Use the empirical ~17% churn base rate as the primary planning prior instead of complex tree models, saving engineering overhead while achieving optimal Brier loss (0.1410).
- **Sensitivity Guardrails:** Model downside scenarios using the 95% confidence interval bound (14.73% churn), which still implies a minimum $321k/array overstatement.

For Customer Success & Operations Leadership:

- **Do Not Intervene Based on Individual Telemetry Scores:** With an ROC-AUC of 0.509, the telemetry does not reliably distinguish churners from renewers at the array level. Punitive interventions will result in high false-positive rates.
- **Expand Predictive Feature Instrumentation:** Collect high-signal leading indicators: executive relationship changes, support ticket escalation sentiment, and competitive hardware bids.
